



1. Brief Introduction

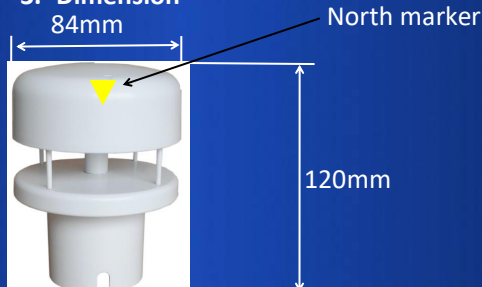
HY-WDC2E Ultrasonic Anemometer is designed to simultaneously measure the 2-dimensional horizontal components of the wind speed and direction based on principle of TOF(time of flight) of ultrasonic sound wave. Low power chip make its power consumption low to 15mA@5VDC. By using ASA shell allows a lighter weight and robust structure. It is widely used in the automotive industry as well as various outdoor applications.

Our weather station are made in molded thermoplastic, also known as ASA, which has high outdoor weather ability. ASA is extremely resistant against UV radiation of sun, it is frost and heat resistant, standing all climatically conditions for over 10 years.

2. Packing List

Ultrasonic anemometer x 1
4 meters cable x 1

3. Dimension



4. Specification

Parameters	Measure Range	Accuracy	Resolution
Wind Speed	0-40m/s(77 knots)	±5%	0.1m/s(0.2 knots)
Wind Direction	0-359°	±3°	1°
Min,Avg,Max,Gust wind data are available at extra cost.			
Threshold of wind speed: 0.1m/s			
Output rate	1 data per second		
Power supply	5-30VDC 15mA@5VDC Sleep mode<1mA@5VDC		
Start up time	< 2 seconds		
Protection Class	IP65		
Operating Temp.	-40℃ to +60℃		
Storage Temp.	-50℃ to +80℃		
Output options	RS232,RS485,SDI-12 4-20mA(has extra cost)		
Color options	White-default or Black		
Optional protocol	MODBUS-RTU,NEMA0183,Active output in ASCII,SDI-12 Only Active output in ASCII support sleep mode.		
Dimension	Φ84×120mm		
Material of body	ABS 0.18kg or aluminium alloy 0.7kg		
Warranty	24 months; Mean Time Before Failure: 10 years		
Certification	CE		

5. Calibration

Each HY-WDC2E ultrasonic anemometer is calibrated within our own wind tunnel to ensure its high accuracy and calibration is based on fundamental physical principles and does not change with use. Recalibration should therefore not be necessary after leaving factory.

6. Communication Interface

We have RS232, RS485, SDI-12 and 4-20mA (has extra cost) for option, interface should be decided when you place order, it can be changed after ex-factory.

7. Communication Protocol

We have MODBUS-RTU, NEMA0183, Active output, SDI-12 for option, protocol should be decided when you place order, it can be changed after ex-factory.

7.1 MODBUS-RTU protocol (default: 9600bps EVEN 8 1)

Inquiry: 01 03 00 00 00 04 44 09

Response: 01 03 08 5D FF 00 22 00 00 40 00 97 BA

Analyze:

wind direction: 00 22 => 34°

wind speed: 00 00 40 00 => 2.0 m/s CDAB float type IEEE754 Standard

Change device address:

Send: 1. >*\r\n 2. >ID 5\r\n 3. >!\r\n

Change baudrate, parity:

Send: 1. >*\r\n 2. >CUS 9600 8-N-1\r\n 3. >!\r\n

7.2 NEMA0183 protocol (default: 9600bps NONE 8 1)

\$WIMWV,180,R,0.5,M,A*32<CR><LF>

wind direction: 180°

wind speed: 0.5 m/s CDAB float type IEEE754 Standard

32:Checksum, the 2 digit Hex Checksum sum figure is calculated from the Exclusive OR of the bytes between (and not including) the \$ and * characters.

7.3 Active output protocol (default: 9600bps NONE 8 1)

:01,56,0.5,28.9,61.1,929.0,12,1,53.60,123.5,21,1,10.1,323,10.324774,103.823456,60.00,10
0,5000.0,200.0,3475.6,12.5,50.3,2.82,11.523,200000.1,0.44125,0.45874,22000,09

wind direction: 50°

wind speed: 0.5 m/s

09:LRC checksum from (include) : to last ,

7.4 SDI-12 protocol (default:1200bps EVEN 7 1)

?! Returns Unit Address (default is 0).

a Current unit address letter (factory default is 0, range is 0 to 9, a to z, A to Z).

b New address letter, range is 0 to 9, a to z, A to Z.

aAb! Change unit address from a to b see above.

aM! Address, Relative Wind Direction, Relative Wind Speed, Corrected Wind Direction, Corrected Wind Speed (GPS and compass option units only), Status.

Command	Description	Response	Example
?!	Unit Address	a<CR><LF>	0<CR><LF>
aAb!	Change the unit address a = 0, the default. b = the new address.	b<CR><LF>	1<CR><LF>
aI!	Unit Identification	allccccccmmmmmmvvvxx...x x<CR><LF>	014HONGYUV 1000002.3000<CR><LF>
aM!	Address, Relative Wind Direction, Relative Wind Speed, Corrected Wind Direction, Corrected Wind Speed (GPS and compass option units only), Status	atttn<CR><LF> a is unit identifier. ttt is time in seconds. n is number of data values. Measurement command to retrieve a reading of the maximum time the anemometer will take to complete a measurement, have data ready and the number of data values	00015<CR><LF> 0 is sensor address. 001 is the specified time, in seconds, until the sensor will have the measurement(s) ready 5 the number of measurement values the sensor will make and return in one or more subsequent D commands; n is a single digit integer with a valid range of 1 to 9 (Direction, Speed, Corrected Direction and Corrected Speed)
aD0!	Retrieve a line of the above data	<address><relative wind direction><relative wind speed><corrected wind direction><corrected wind speed><status><CR><LF>	0+056+00.50+012+02.82+1111<CR><LF> Status indicate the validity of each bit, 0-invalid, 1-valid In this case, Status 1111 1 - +056 valid wind direction 56° 1 - +00.03 valid wind speed 0.5m/s 1 - +012 valid 1 - +02.82 valid

Typical Applications

- Highway
- Air Pollution
- Smart city weather station
- Agriculture
- Fire weather monitoring
- Bridge monitoring
- Roadside smart lamp

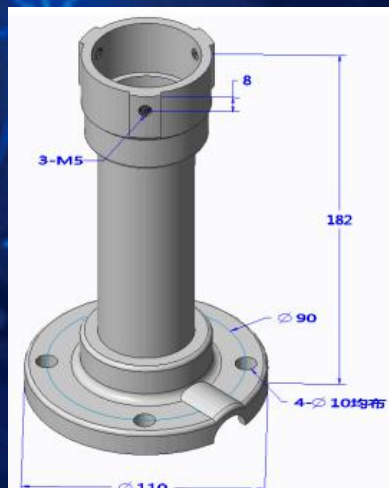


Mounting Kits 1



25 USD/UNIT

Mounting Kits 2



25 USD/UNIT